Type MS · Calculate and express a ratio in a required form

Answer 1 of 11 · Question 5 part a.i · 2 marks

2024 NOV · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

5 (a)	(i)	• 13 750 ÷ 7500; • 1:1.83;	Allow 1:1.8/1:2/ 1:1.8333... allow if they do it the other way round ie 7500 ÷ 13750	2
	(ii)	• person A has an infection/disease; • Immune response/more white blood cells needed, to produce antibodies/phagocytosis/destroy pathogen/virus/bacteria;	ignore sick ignore fight infection allow alternative correct explanation	2
	(iii)	Any two from: • excess bleeding; • poor/no blood clotting; • poor/slow wound healing/more time for pathogens to enter wound;		2
(b)	(i)	• tiredness/fatigue/high heart rate/faster breathing; • less oxygen carried/transported (to body cells); • less aerobic respiration; • less energy;		4
	(ii)	• HA is dominant; • parents are heterozygous; • HA expressed in heterozygous form; • so only needs one allele;	ignore ref to allele/gene for mpt 1	4

Total Marks 14

Type MS · Calculate and express a ratio in a required form

Answer 1 of 11 · Question 5 part a.i · 2 marks

2024 NOV · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Calculate and express a ratio in a required form

Answer 2 of 11 · Question 7 part a.i · 2 marks

2020 JAN · Paper 01 · find part a.i in this mark scheme extract

number				
7 (a)	(i)	• 67.5 ÷ 48.3 (1) • 1:1.4 (1)	Full marks for correct final answer	2
	(ii)	Any two from • more males than females are overweight in all countries (1) • lowest percentage of people between the ages of 18-24 in all countries are overweight (1) • increased in all countries 18 - 64 (1)		2
	(iii)	sample size too small (1) not all overweight people identified/sample not representative (1)		2
(b)	(i)	Any three from: • pancreas (1) • releases insulin (1) • reference to the liver taking up glucose (1) • glucose converted to glycogen (1)	R sugar R breaks down glucose	3
	(ii)	Any two from: • high blood glucose increases the concentration of the blood (1) • water moves out of cells (1) • by osmosis (1) • excess water in blood excreted in urine (1)		2

Total for Question 7 = 11 marks

Type MS · Calculate and express a ratio in a required form

Answer 3 of 11 · Question 6 part a.ii · 2 marks

2020 JAN · Paper 02 · find part a.ii in this mark scheme extract

Number				
6	(a) (i)	• more males than females die (from *S.aureus* and MRSA)(1) • number of deaths from both *S.aureus* and MRSA increase with age(1)	Allow other valid conclusions e.g more deaths from both microorganisms in males/females aged 85 and over	2
	(ii)	416.6 ÷ 1.8(1) 231.4:1 (1)	Allow 231:1 Allow full marks for correct final answer	2
	(iii)	41.2 + 21.9 = 63.1(1) 53 × 63.1 = 3344(1)	Allow 3344.3 Allow full marks for correct final answer	2
	(iv)	Any two from: • MRSA is resistant (1) • to (many) antibiotics(1) • (MRSA infection) difficult to treat(1) • reference to MRSA infection being (highly) contagious/out of control/easily transferred(1)		2
	(b) (i)	mircroorganisms/bacteria/fungi	Ignore viruses Ignore named antibiotics	1
	(ii)	Any six from: • sterilise Petri dishes/agar/nutrient broth(1) • obtain sample(1) • sterile inoculating loop/sterile swab(1) • pass neck of flask containing culture through a flame(1) • smear culture/sample/swab onto agar(1) • reference to adding antibiotics (to agar)(1) • seal Petri dishes/cover with lid(1) • incubate(1) • measure area of clear zone around antibiotic discs(1)	Allow description of how loop/swab is sterilised	6

Total for Question 6 = 15 marks

Type MS · Calculate and express a ratio in a required form

Answer 4 of 11 · Question 1 part c.i · 2 marks

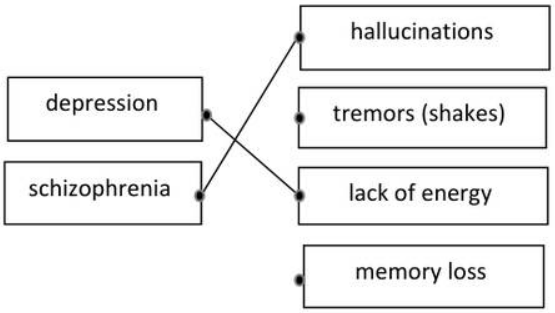
2020 NOV · Paper 01 (regional R) · find part c.i in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Calculate and express a ratio in a required form

Answer 5 of 11 · Question 2 part a.iii · 2 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part a.iii in this mark scheme extract

2 (a) (i)	2002;		1
(ii)	2005 (1.4)		1
(iii)	2.75:2.6; 1.1/1.06 :1;		2
(iv)	low income/low self esteem/(domestic) violence/low social status/substance abuse;		1
(b)			2
(c)	**A** alcohol and heroin; B antibiotics have no effect on mental health C paracetamol has no effect on mental health D antibiotics have no effect on mental health		1
(d)	Any two from the following brain/skin/sweat glands/muscles/blood vessels/arterioles;;		2
			Total 10

Type MS · Calculate and express a ratio in a required form

Answer 5 of 11 · Question 2 part a.iii · 2 marks

2022 JUN · Paper 02 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Calculate and express a ratio in a required form

Answer 6 of 11 · Question 4 part c · 2 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

Question number				
4	(a)	• correct plots;;; • key/labels;		4
	(b)	• correct components/nutrients/list of components min of five; • in correct amounts/proportions;		2
	(c)	• 75; • 15 • = 5:1;	ECF	2
	(d)	• contain protein • needed for growth/repair/,muscles/tissues; • area A girl has low(er) levels/area B girl has high(er) levels; • less/poor growth/area B girl good growth; • kwashiorkor;		5

Total 13 marks

Type MS · Calculate and express a ratio in a required form

Answer 6 of 11 · Question 4 part c · 2 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Calculate and express a ratio in a required form

Answer 7 of 11 · Question 3 part a.i · 2 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

Number				
3	(a) (i)	250:750; 1:3;	full marks for correct answer with no working	2
	(ii)	17 500 - 5600 = 11900; $\frac{11\,900 \times 100}{5\,600}$; = 210%;	full marks for correct answer with no working 213/212.5 = two marks	3
	(iii)	• other parts of the body/organs; • liver/kidney/lungs/arteries/veins; • needs some blood supplied/blood flows to these organs;		3
	(b) (i)	• large/big/huge increase in total flow; • large/big/huge increase in flow skeletal muscles/skin; • decrease for gut;		3
	(ii)	• heat generated(body) temperature increases during exercise; • from respiration; • carried by blood/more blood to skin; • (heat) lost through skin/by radiation;		4

Total 15 marks

Type MS · Calculate and express a ratio in a required form

Answer 7 of 11 · Question 3 part a.i · 2 marks

2023 NOV · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Calculate and express a ratio in a required form

Answer 8 of 11 · Question 4 part c.i · 1 mark

2023 NOV · Paper 02 · extract 1 of 2 · find part c.i in this mark scheme extract

4 (a)	• reduce fat/cholesterol intake; • increase exercise;		2
(b) (i)	Any four from • liver breaks down toxins/drugs; • (if not working properly) statins won't be broken down/will accumulate/reduced detoxification; • kidneys excrete waste (products)/named waste products; • (if not working) can't be removed/less waste/named waste filtered/removed; • increased risk of (further) liver/kidney damage;	Ignore references to deamination	4
(ii)	• statins could cross from mother to fetus; • through the placenta; • damage/affect growth/development of fetus/lead to low birth weight of fetus; • could be in mother's milk if breastfeeding;	Allow baby for mp's 1 and 3 Ignore affects fetus	3
(c) (i)	• (16.3 : 8) = 2.0;	Accept 2.0375 Accept 2/2.04	1
(ii)	Any four from • Double the number/more symptoms shown with patients taking statins than those taking no tablets • even on statins symptoms not high; • almost same with placebo/symptoms with placebo similar/just below those on statins; • symptoms still present when taking no statins; • statins not the cause/other causes of symptoms/named cause e.g. age/gender/fitness/lifestyle;		4 exp

Total 14 marks

Type MS · Calculate and express a ratio in a required form

Answer 8 of 11 · Question 4 part c.i · 1 mark

2023 NOV · Paper 02 · extract 2 of 2 · find part c.i in this mark scheme extract

Question number	Answer	Notes	Marks
			2

Type MS · Calculate and express a ratio in a required form

Answer 9 of 11 · Question 6 part b.ii · 3 marks

2024 JUN · Paper 02 (regional R) · find part b.ii in this mark scheme extract

6	(a)	• reduced blood flow to the heart/cardiac muscle/coronary arteries; • less oxygen to heart/cardiac muscle/tissue; • less (aerobic) respiration/less energy for contraction; • (cardiac) muscle/cells die; • (leading to) heart attack/heart disease;		(4)
	(b) (i)	As the use of statins increases the level of blood cholesterol decreases;	allow inversely proportional	(1)
	(ii)	1988 = 31; 2005 = 17; 1 : 1.8/1 : 0.54/1:0.548/1:0.55;	ecf for incorrect graph readings. if use 18:17 ratio is 1:1.0588 (accept 1:1.059, 1.06, 1.1) full marks for correct final answer	(3)
	(iii)	• select sample of people; • record the number of people taking statins/not taking statins; • take blood samples; • measure the concentration/level of cholesterol (in blood); • repeat every year/between 1988-2008;		(5)
	(iv)	any two from the following: • gender/age not taken into account; • sample size maybe too small; • reference to a named lifestyle not taken into account e.g. diet, level of activity/smoking; • genetic predisposition/diseases not taken into account;		(2)

Total for Question 6 = 15 marks

Type MS · Calculate and express a ratio in a required form

Answer 10 of 11 · Question 4 part c.iii · 3 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part c.iii in this mark scheme extract

4 (a)	• diarrhoea by clean water/safe water/ bottled water/boiled water/ sewage treatment/chlorination/good sanitation/not mixing sewage with drinking water/washing hands/good hygiene/ cooking food thoroughly; • malaria by nets/draining swamps/insecticides/ insect repellent's/wearing clothing to cover arms (and legs/mosquito coils/anti-malarial/vaccination ;	Ignore ORT	2
(b) (i)	antibiotics;		1
(ii)	• elderly have weakened/lowered immune system/immunocompromised; • indoor spaces makes transmission easier/reduced air flow/ reduced ventilation/build- up of air droplets with pathogens/ people closer together;		2
(c) (i)	• diet; • exercise; • smoking: • drinking alcohol: • stress	Accept drugs	2
(ii)	• $\frac{5\,000\,000}{100\,000} = 50$; • $50 \times 445 = 22\,250$; • 22 300; ***22,250 scores 2 marks*** ***22,300 scores 3 marks***	***1 mark for ×445 if there are no other marks***	3
(iii)	• use of 445 and 185; • $\frac{445}{185}$; • 2.41:1/2.405:!/2.4:1= 3 marks;	ecf from mp1 **if they have done the correct calculation and 2:1 is on the answer line they score** 2 marks	3
(iv)	• people live longer/elderly population; • these diseases are more common in older people; • less likely to die younger from infectious diseases/better medical facilities/treatment; • different diet/ more meat/saturated fats/processed food; • drinking more alcohol;	People live longer so there is more cancer and dementia/these diseases= 2 marks	2

Total 15 marks

Type MS · Calculate and express a ratio in a required form

Answer 10 of 11 · Question 4 part c.iii · 3 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part c.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Calculate and express a ratio in a required form

Answer 11 of 11 · Question 4 part a.ii · 2 marks

2025 JUN · Paper 02 (regional R) · find part a.ii in this mark scheme extract

4	(a)	(i)	• decreases; • from 82% to 72%/by 10%		2
		(ii)	79:21; 3.76:1;	accept 3.7612:1 accept 3.762:1 accept 3.8/4:1	2
		(iii)	any two from • passive smoking/ex- smokers already have damaged lungs; • (air) pollution; • genetics/heredity;		2
		(iv)	tar;		1
	(b)	(i)	bronchiole;		1
		(ii)	alveolus;		1
		(iii)	any four from • damaged alveoli/alveoli walls broken; • less surface area (for gas exchange); • difficulty breathing/shortness of breath; • less oxygen passing into the blood/ less carbon dioxide removed from blood/less gaseous exchange; • reduced (aerobic) respiration; • fatigue/less energy/not very active;		4
				Total for Question 4 = 13 marks	